The Terminal Architecture

The Dream-Paper Suite: Final Status and Honest Frontiers
Papers A through BV (75 Papers)
A Proposed Paper BV

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March 20, 2026

Abstract

The dream-paper suite spans 75 papers (A through BV), comprising the finite relational kernel, the Yang–Mills Millennium Theorem, the Unified Emergence Theorem, thermodynamics and causality, matter representations, Lorentzian geometry, Born rule, spacetime dimensions, EFE structural form, and the complete NS closure chain.

The suite's architecture divides into three tiers:

Tier 1 — Fully proved (Unc): Yang–Mills mass gap, gauge group $G = SU(3)\times SU(2)\times U(1)$, three matter generations, Born rule, 3+1 spacetime dimensions, Leray weak NS existence, causal order, entropy, time, matter representations. All from the finite relational kernel alone.

Tier 2 — Conditionally proved (Cond) under named hypotheses: NS smoothness (H-CODIM2 + H-LR), Lorentzian geometry (Phase-A), EFE structural form (Phase-A + KPO-3), Born rule (Phase-A + PASS).

Tier 3 — Open, precisely formulated: H-LR (Lieb–Robinson for fluid sector), H-CODIM2 (interior rate decay $\eta_n \to 0$, the Renormalization Flow Conjecture), EFE exact coefficients (gravity NS-Stage-3).

The NS Prize question has been reduced to two hypotheses that are internally motivated, precisely stated, and addressed to specific objects in $\mathfrak{A}_\infty$.

1 The full program in eight layers

Layers 1–6: A through BB (Yang–Mills) — all Unc

Papers	Program	Status
A–G	Finite kernel, LS-2, capacity, certificates, sector, YM bridge, NS bridge	Unc/Cond
AA–AK	POT category, $\mathcal{U}$, $T_F \hookrightarrow K^* \hookrightarrow \mathcal{U}$	Unc
AL–AY	Continuum: $\mathfrak{A}_\infty \cong \mathcal{A}(M^4)$	Unc
AN–AQ	Connection $(\mathcal{E}, \nabla)$, $D^*F = 0$, $\mathfrak{g}_\infty > 0$	Unc
AZ–BB	KMS vacuum, quantum gap $\Delta > 0$, **Yang–Mills Millennium Theorem**	**Unc**

Layer 7: BC–BE (Thermodynamics, Causality, Unification) — all Unc

Ledger–Entropy Architecture (BC): time and entropy unified from one balance law. Causal Reconstruction (BD): Lieb–Robinson causal preorder. **Unified Emergence Theorem** (BE): all physical

layers are derived shadows of $\mathcal{U}$.

Layer 8: BF–BV (Matter, Geometry, NS, Unification)

Paper	Content	Status
BF	Canonical matter reps (adjoint + fundamental)	Unc
BG	Lorentzian geometry from causal preorder (HKM)	Cond (Phase-A)
BH	EFE structural form + coefficients (Thm 41.15, 41.17)	Cond (Phase-A, KPO-3)
BI	NS quasi-decay via LR (Paper G update)	Cond (H-LR)
BJ	9→4 Dimensional Selection	Cond → Unc (via BL/BO)
BK	Final master ledger (A–BJ)	Unc (architecture)
BL	Born rule + N-add/T-add/BV for barbell	Unc (Born: Cond)
BM	Three generations from three dS_v-blocks	**Unc**
BN	The Maximal Honest Claim	Unc (architecture)
BO	WL-1 discharge; spacetime 3+1 fully Unc	**Unc**
BP	EFE complete: Λ finite-volume effect, diff. inv.	Cond
BQ	NS structural program: Stages 0–2, Leray weak existence	Unc/Cond
BR	Unified Coercive-Transport Inequality (YM/NS/grav)	Unc
BS	8-step NS closure chain (H-LR ⇒ smooth)	Cond (H-LR)
BT	Codim-2 concentration + Quadratic Floor (H-CODIM2+H-LR)	Cond
BU	Interior Stability Barrier: $\eta_n = 2/3$ (current axioms)	Unc (barrier)
BV	This architecture summary	Unc (architecture)

2 The NS Prize frontier — precisely

Hypothesis	Statement	Source
H-LR	LR bound applies to fluid-observable subalgebra of $\mathfrak{A}_\infty$	Paper BD (full LR: Unc)
H-CODIM2	Interior certified-event rate $\eta_n = O(K_0^{-n/2})$	= Conj. 36.28 + H-LR
NS Prize	u globally smooth under H-CODIM2 + H-LR	Cond (Papers BS, BT)
Current status	$\eta_n \approx 2/3$ (constant, not decaying)	Theorem BU1 (Unc)

The NS Prize question in NFC language:

Does the Lieb–Robinson propagation in $\mathfrak{A}_\infty$, restricted to the fluid-observable sector, induce an effective dynamical renormalization that suppresses the interior certified-event rate from $\sim 2/3$ to $O(K_0^{-n/2})$?

3 The three remaining open obligations

Obligation	What it requires	Character
H-LR	Fluid sector inherits LR bound	Sub-algebra identification
H-CODIM2	Effective rate decay $\eta_n \to 0$	Dynamic RG on collar walk
EFE exact coefficients	$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ numerically	Gravity NS-Stage-3 analogue

Remark 3.1 (On the nature of the remaining obligations)**.** *All three are precisely stated, addressed to named objects in $\mathfrak{A}_\infty$, and internally motivated by the suite's architecture:*

H-LR*: The full Lieb–Robinson bound on $\mathfrak{A}_\infty$ is* Unc*. H-LR asks only which part of L_∞ generates fluid dynamics — an operator identification question, not a new structural principle.*

H-CODIM2*: The monograph explicitly names this as requiring a "dynamical renormalization group acting on the collar walk itself." The Lieb–Robinson decay suggests a candidate mechanism: at scale a_n, Lieb–Robinson exponentially suppresses far-field collar-break contributions, giving effective rate $\eta_n^{\text{eff}} \approx \frac{2}{3} e^{-\mu a_n} \to 0$. The obstacle is that this heuristic has not been formalized as a theorem inside the NFC axiom set.*

EFE exact coefficients*: Structural EFE form is proved (Paper BH, Cond). The exact numerical coefficients require an L^2_{loc} convergence theorem for the discrete interface equation — the gravity NS-Stage-3 analogue.*

None of these is a conceptual gap. All three are identifiable theorem tasks whose difficulty is the NS Prize difficulty (for the first two) and a technical scaling analysis (for the third).

4 The suite in one sentence

The dream-paper suite derives, from a finite relational kernel and without importing any physical or geometric primitive, the gauge group, matter generations, quantum mass gap, Born rule, 3+1 spacetime dimensions, Leray NS existence, and the complete structural skeleton of gauge dynamics, thermodynamics, and causality — all as derived shadows of the universal persistent observational-transfer object $\mathcal{U}$ — while identifying the NS Prize question as the question of whether the Lieb–Robinson propagation in $\mathfrak{A}_\infty$ induces an effective dynamical renormalization of the NFC collar walk.